

Physics is the branch of science that studies how nature behaves. It relates to all other types of sciences: chemistry, biology, geology, astronomy, mechanics, etc. Since the knowledge that we have about the behavior of nature is constantly growing and changing, so is this branch of science.

FORCE and MOTION

Force is any interaction that, when unopposed, will change the **motion** of an object. Force can cause an object with mass to change its velocity (which includes to begin moving from a state of rest), i.e. to accelerate. Force can also be described intuitively as a push or a pull. The SI unit of force is newton (N). Its SI base units is kg-m/s². Other units are dyne, poundal, pound-force and kip.

$$\begin{aligned} 1 \text{ N} &= 1 \text{ kg} (1 \text{ m/s}^2) \\ 1 \text{ dyne} &= 1 \text{ g} (1 \text{ cm/s}^2) \\ 1 \text{ lbf} &= 1 \text{ slug} (1 \text{ ft/s}^2) \end{aligned}$$

Motion is the phenomenon in which an object changes its position over time.

Inertia is the property of matter that causes it to resist a change in its state of motion. *Excel Review Center*

SCALAR and VECTOR

Scalar quantity is a quantity that has only **magnitude**. *Excel Review Center*

Examples: Length, Area, Volume, Speed, Distance, Mass, Density, Temperature, Pressure, Energy, Work, Entropy, Power, Resistance, Charge, Mechanical Advantage, Electric and Magnetic Potential, Angle, etc.

Vector quantity is a quantity that has both a **magnitude** and a **direction**.

Examples: Displacement, Direction, Velocity, Acceleration, Momentum, Impulse, Force, Electric Field, Magnetic Field, Weight, Torque, Moment of Force, Dipole moment, etc.

Newton's Laws of Motion

First Law

(Law of Inertia)

"A body at rest will remain at rest, and a body in motion will remain in motion unless it is acted upon by an external force."

The ball will no move unless there is a force to cause it to move.

If a ball were rolling on a frictionless surface, it would keep moving unless met with an outside force.

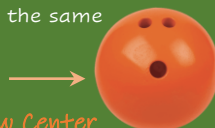


Second Law

(Law of Acceleration)

"The force acting on an object is equal to the mass of that object times its acceleration." $F = ma$

Because the mass of each ball is different, each ball will travel a different distance and at a different speed when it is hit with the same force.



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Third Law

(Law of Reaction)

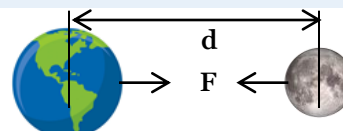
"For every action, there is an equal and opposite reaction."



When there is a force by one side, there will be opposite and equal force by the other side, cause each to move in opposite direction.

Newton's Law of Universal Gravitation

"The gravitational force between two masses is directly proportional to the product of the two masses (m_1 and m_2) and inversely proportional to the square distance, d , between them."



$$F = G \frac{m_1 m_2}{d^2}$$

G = gravitational constant = $6.673 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$

Work, Power and Energy

Work is the product of the force applied to an object and the displacement through which the force acts on an object.

$$W = F \times d \quad \text{or} \quad W = Fd \cos \theta \quad (\text{if displacement makes an angle } \theta \text{ with the force})$$

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The standard unit of work and energy is joule (J).

$$1 \text{ J} = 1 \text{ N-m} = 1 \text{ kg-m}^2/\text{s}^2$$

$$1 \text{ erg} = (1 \text{ dyne})(1 \text{ cm})$$

$$1 \text{ ft-lb} = (1 \text{ lbf})(1 \text{ ft})$$

Energy is the capacity to do work. **Potential energy (PE)** is the energy in a body due to its **position**. **Kinetic energy (KE)** is the energy in a body due to its **motion**.

$$\begin{aligned} \text{KE}_{\text{translational}} &= \frac{1}{2}mv^2 \\ \text{PE} &= mgh \end{aligned}$$

$$\begin{aligned} \text{KE}_{\text{rotational}} &= \frac{1}{2}I\omega^2 \\ v &= r\omega \end{aligned}$$

Where: I = rotational inertia of an object, ω = angular velocity, v = linear velocity

Power is the rate at which work is done or the rate at which energy is expended. Unit of power is watt (W). Another unit of power is horsepower (hp).

$$1 \text{ W} = 1 \text{ J/s} = 1 \text{ N-m/s}$$

$$1 \text{ hp} = 746 \text{ W} = 550 \text{ ft-lb/s}$$

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